Evaluation of Barite

1) Density by Le Chateller flask

1- Fill a Le Chateller flask with kerosene below the zero mark and let it to expand in water bath with constant temperature 32^0 C for at least 1 hr.
   - **If the kerosene level is outside the -0.2 ml to +1.2 ml volume range, use 10 ml pipette to add or remove kerosene to bring it within range.**
   - **With tissue paper, wipe diagonally along the length of the dowel.**

2- Keep eyes at meniscus level and record the initial volume v_1
3- Weight 80.0 g ± 0.05 g of ***dried*** barite and transfer it to the flask carefully, record m
4- Return the flask to the bath and let stand for at least 0,5 hr.
5- Record the final Volume v_2
6- **Calculation – Density by Le Chateller flask**

$$\rho = \frac{m}{v_2 - v_1}\ (g/ml)$$

2) Water-soluble alkaline earth metals, as calcium

1- Weight 100 g ± 0.05 g of barite and transfer to an Erlenmeyer flask then add 100 ml D.W.
2- Stopper the flask and shake for 30 min using mechanical shaking apparatus.
3- Filter the suspension using two filter papers on LPLT W.L and collect the filtrate.
4- Take 10 ml of the filtrate into titrating vessel + Vers. Buffer + Vers. Indicator ≠ 0.02 N EDTA. Record as v_4
5- Record $V_{\text{end. point}}$ of EDTA as v_3
6- **Calculation – Water soluble alkaline earth metals**

$$m_1 = 400\left(\frac{v_3}{v_4}\right)\ (mg/kg)$$

3) Residue of diameter greater than 75 & 45 µm

1- Weight the sieve and record as m_1
2- Weight 50 g ± 0.05 g of ***dried*** barite and add to 350 ml water containing 0.2 g Sod. Hexametaphosphate, stir for 5 min. ± 1 min. (Hexametaphosphate is not necessary & 50 g of sample)
3- Placed into sieve and wash with tap water.
4- Dry the sieve in the oven and weight the sieve and record as m_2
5- Calculation – Residue of diameter greater than 75 and 45 µm

$$w_1 = 100\left(\frac{m_2 - m_1}{m}\right)\ (\%)$$

4) Particles less than 6 µm in equivalent spherical diameter

- **Hydrometer correction**:
  Place a cylinder which contains water in a constant temperature bath, then record the reading of hydrometer for lowest and highest temperature could be obtained and take a thermometer reading (approx.18.0 & 25.0 °C).

  Record the hydrometer reading as R_1 and R_2 and the temperature reading as θ_1 and θ_2 then use the sheet of calculations "% Particle less than 6 mic".

- **Preparation of dispersant Solution**:
  40.0 g ± 0.1 g of sodium hexa-metaphosphate and 3.5 g ± 0.1 g anhydrous sodium carbonate -to adjust pH=9- for 1000 ml solution.

1- Weight 80 g of ***dried*** barite (m) then add 125 ml ± 2 ml (127 g ± 2 g) of dispersant solution and add approx. 400 ml D.W and stir for 5 min.
2- Transfer to the sedimentation cylinder and add D.W to the 1000 ml mark, close the cylinder with rubber stopper and mix well -upright to the inverted position and back- for 1 min. to obtain a homogeneous suspension.
3- Put the Cylinder into water bath and simultaneously start the timer and hang the thermometer in the water bath.
4- Put the hydrometer slowly to approximately 1.020 reading before releasing.
5- Record The Hydrometer readings at (10 min, 20 min, 30 min, 40 min ……) or until the first point below the 6 µ value is reached

   Note: read the top of the meniscus at the required time, then remove the Hydrometer immediately slowly and clean after each reading and rinse the hydrometer with D.W.

6- Record the time t (in minutes), the temperature θ (in ^{0}C or ^{0}F), and Hydrometer reading R, then use the sheet of calculations "% Particle less than 6 mic".
7- **Calculation - Particles less than 6 µm in equivalent spherical diameter**

Barite _ EMEC Bar\% Particle Less than 6 mic. calc.xlsx

5) Iron Content

Chemical and physical Requirements

Test Method	PARAMETER	REQUIREMENT
API	Density, g/ml	Minimum 4.20
API	Water-soluble alkaline earth metals as calcium, mg/kg	Maximum 250
API	Residue greater than 75 µm, mass fraction %	Maximum 3.0
API	Particles less than 6 µm in equivalent spherical diameter, mass fraction %	Maximum 30.0
EMEC Standard	Iron Content	